

Classical $\text{KL}(p\|q) := \sum_i p_i \log \frac{p_i}{q_i}$

$$|x|_1 := \sum_i |x_i|$$

$$\text{KL}(p\|q) \geq \frac{1}{2} |p - q|_1^2$$

Quantum $\text{KL}(P\|Q) := \text{Tr}(P(\log P - \log Q))$

$$|X|_* := |\text{svd}(X)|_1$$

$$\text{KL}(P\|Q) \geq \frac{1}{2} |P - Q|_*^2$$



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